

Today we worked together to complete the kronig-penney model calculations. As a group we agreed on using a mathematica notebook to calculate our transcendental equations and were able to extend this to getting plots of allowed energy states and therefore the energy gaps like figure 8.7 in Desai. The mathematica plot also allows for us to vary the parameters such as the potential and the well width/barrier width. We also attempted to plot the wavefunctions themselves. These calculations and figures are in the attached mathematica files and we encourage you to play around with the parameters. As a great man once said, the world is your oyster. And in this case, the mathematica notebook would be considered the oyster.